From Persistent Disagreement to Agreement: Learning with Revisions of Perceptual Sets

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ABSTRACT

We often view sensation and perception as one process and mistakenly take our internal representation of the physical world as the world itself. Cognitive biases could occur in the perception process without our noticing. This paper zooms in on the learning process and highlights a specific subprocess—the perception process. Agents take the same sensory information as inputs, direct them through different *perceptual sets* and thus form different *perceptual information*, which is then used to update posterior beliefs. Biased perceptual sets thus lead to persistent disagreement among agents. A revision of the perceptual set can generate a discrete revision of posterior beliefs, even when no new raw information arrives. Full revisions align beliefs with the common-perception Bayesian benchmark; partial revisions narrow disagreement without necessarily eliminating it.

Keywords: perceptual bias, cognitive frictions, persistent beliefs

JEL codes: D82, D83, G11, G14, G41.

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1 Introduction

The world overflows with information that is visual, auditory, olfactory, gustatory, and etc. In order to make sense of all this *external* information, our body has developed a series of processes for sensing and perceiving the outside world. The first step is *sensation*, "the process of detecting external events with sense organs and turning those stimuli into neutral signals" (Krause et al., 2017). The sensory information is then relayed to the brain where the second step, *perception*, occurs. Perception is "the organization, identification, and interpretation of sensory information" (Schacter et al., 2014) in order to form an *internal* representation of the world. As an example, when investors are flipping through business news in the *Wall Street Journal*, the sensory receptors in their eyeballs are registering different patterns of light reflecting off the pages, converting them to neutral signals which are then integrated and processed by the brain into meaningful perception of words. However, humans are not machines; what is conveyed physically on the paper is likely not what readers understand.

According to Bruner and Postman (1949), when encountering new information, we are open to different cues. However, as we process more information, some cues become familiar and thus selective. We search for more cues that confirm the categorization of the target. We also actively ignore and even distort cues that violate our initial perceptions. Our perception becomes more selective and we finally re-create a consistent internal presentation of the new information. In other words, we interpret patterns of stimuli through a *perceptual set*—"a readiness to perceive a stimulus in a particular way" (Weiten, 2001).

An important factor that heavily influences our perception is our expectation (Bruner and Minturn, 1955), which is characterized by the prior belief in this chapter. Perceptual sets are also affected by motivational forces—people have a tendency to see what they want to see (Balcetis and Dunning, 2006), and the selection of sensory input—people tend to perceive what they focus their attention on (Chun and Wolfe, 2005) and overlook unexpected items and events that do not receive their attention (Simons and Chabris, 1999). Another factor that affects perception sets

is *confirmation bias*, that is people seek information that supports their prior beliefs and ignore information that contradicts them (Nickerson, 1998). Confirmation bias is also widely studied in the finance literature.

Motivational factors, selective perception and confirmation bias all vary from person to person; it is akin to "individual fixed effect" and thus summarized in this chapter by the blanket term *personal traits*. These personal traits make perception sets very persistent—it is hard for people to change how they view the world. Nevertheless, it is not completely impossible to achieve that, especially after disastrous events. Those events are so intense that people's attention is drawn to them even if it was placed elsewhere and that contradictions to prior beliefs is too significant to overlook. In other words, disastrous events post *perceptual shocks* that suddenly and greatly change perceptual sets.

Individuals process the same *sensory information* through different perceptual sets and thus form different *perceptual information*, that is different internal representation of the same external inputs, and then combine them with the prior belief to form posterior beliefs[1]. The processes of perception and belief updating are shown in Figure 1.

To model dynamic belief updating and jumps of perceptual sets, I introduce the tools that are frequently used in continuous-time finance. Models of utility in stochastic continuous-time settings typically assume that beliefs are represented by a probability measure. For example, Chen et al. (2024) models two competitive agents, one with the objective belief in the sense that her probability measure is consistent with the physical measure, and the other with a probability measure that is different from but equivalent to the physical measure. In this chapter, I model perceptual sets as probability measures, while beliefs about the unobservable process are obtained through Kalman filter. Individuals are not aware of the existence of perceptual sets; how the external information is converted to internal information is a blackbox to them. Starting from a common prior, individuals observe a random process $\{P_t\}$ and try to make inferences about a

[1]In psychology, belief is defined as the mental acceptance of something to be true (Weiten, 2001), which is consistent with its economic interpretation as a probability distribution over a set of possible states.

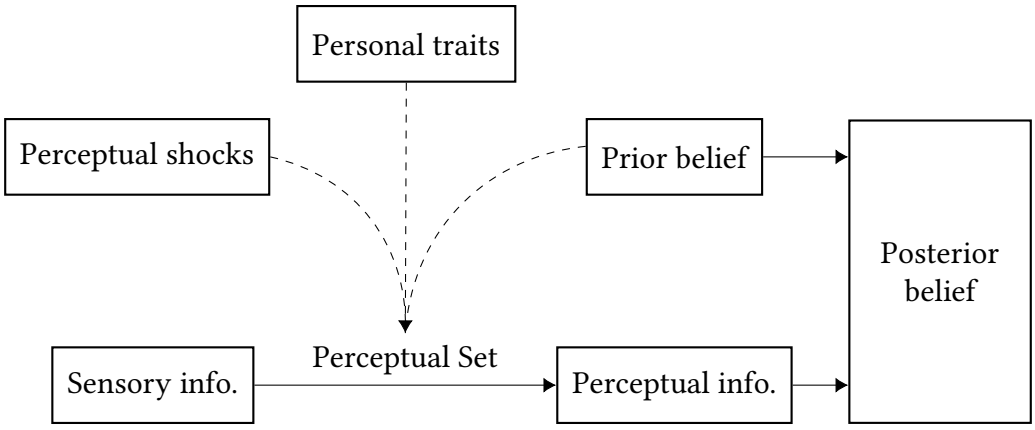


Figure 1. Sensory information, perception and belief. I abstract from sensation and directly take sensory information as raw input. It goes through the perception process, during which it is filtered by a subjective perceptual set and converted to *perceptual information*. It is then combined with the prior belief to form the posterior belief. The perceptual set is determined by personal traits (motivational factors, selective perception and confirmation bias), the prior belief and perceptual shocks. Perceptual shocks could make the perceptual set subject to sudden and big changes.

function of P_t and an unobservable random process $\{\theta_t\}$. The inference is de facto done under individuals' subjective probability measures. In the spirit of entrenchment for beliefs[2], I define a notion of *set entrenchment*. A perceptual set with low entrenchment is more subject to revision. Agents are endowed with an initial set entrenchment level. Arrivals of independent perceptual shocks and shocks from P_t will affect entrenchment. The perceptual set is revised when the entrenchment level hits the set revision threshold. Even with common prior and information, individuals have persistently different beliefs. In somes cases, we also observe side-switching.

Another interesting question to think about is to disentangle the effects of new information on the posterior belief versus changes in the posterior belief resulted from perceptual set revision. The first one is the standard Bayesian updating, while the second one has the flavor of Collin-Dufresne et al. (2016), which implemented parameter learning for model selection. This chapter takes a different angle, in that independent perceptual shocks could lead to perceptual set revision, resulting in a different posterior despite the same old information. New information could be absent in this setting.

This general theoretical framework of subjective perceptual sets could be applied to different

[2]In belief contraction, the beliefs with the lowest entrenchment should be the ones that are most readily given up (Hansson, 2017).

areas in economics and finance. We can introduce it to the financial markets to explain excess volatility and trading volume. It also explains why investors initially underreact to certain news but suddenly the market perspective is changed, and why tail events happen more often than common wisdom suggests. On the micro-level, if investors are aware of the existence of the perceptual set but do not know how it is biased, what can they do to improve their utility? And can a social planner devise any policy to increase investors' utility and stabilize the market? The framework can also be used to explain belief polarization in political economy and bubbles in household expectations.

Literature There is a growing literature that tries to explain persistent disagreement among agents. The starting point of many studies is a well-known result by Savage (1954). He argued that repeated observation of signals will lead a Bayesian agent to assign probability 1 to a true event almost surely. What we observe, however, is frequently at odds with this standard theoretical framework; persistent disagreement among agents still exists despite abundant information.

To explain this observation, some research has been done under the rational inattention framework. For example, in Nimark and Sundaresan (2019), agents can choose channels, which are noisy, costly, and have different precisions in different states of the world, to observe the reality. They allocate more precision to those states a priori more likely, which perpetuates differences in beliefs. Kominers et al. (2018) study Bayesian agents for whom computing posterior beliefs is costly; such agents face a tradeoffs between economizing on attention costs and having more accurate beliefs. Even small processing costs can lead to significant departures from the standard costless processing model.

However, humans are not machines; we process information in different ways than a machine does. Psychologists and cognitive scientists have shown that individuals display biases in their perception of the world, upon which another strand of research in economics is based. In that literature, it is often assumed that agents do not know a certain state of the world *and* the quality of information regarding the state. For example, Cheng and Hsiaw (2022) drew on the *curse of*

knowledge, that is individuals who have seen information often fail to ignore it when the context requires using only prior beliefs, to model *pre-screeners* who "double-dip" the data. Pre-screeners first form a first-stage belief about credibility based on newly-observed signals, and then use it instead of their prior to weight all signals. This proposed mistake predicts when individuals over- or underreact to new information, depending on the order in which they received previous signals, and explains why information that should objectively mitigate disagreement may amplify it instead. In a similar vein, Koçak (2018) showed that if a signal is informative on multiple dimensions, and if individuals can hold only a limited number of items in working memory, that is after the agent updates the first belief, she has to discard the prior of the now updated parameter, and use its posterior and the new information to update the second parameter, then the order with which beliefs are updated across different dimensions leads to different posteriors. This provides an underlying mechanism of why the curse of knowledge comes into effect.

In contrast, in this chapter individuals do not doubt the credibility of experts but process sensory information in an *unnoticeably* biased way. Individuals do not question the quality of information source, nor do they make active information acquisition. They are given a full set of information, but each individual process information under a different perceptual set, resulting persistent disagreement. The perceptual set is not just a result from experiences or prior information, but could also be biological, e.g. sensitivity to different types of sensory information. Thus individuals can be free of double-dipping the data but still updating information in a biased way.

In Chauvin (2023), a dissonance-reducing learner wishes to agree with his sources of information, and therefore distorts his subjective perception of the arguments before adding them to the prior. Distortion is achieved through scaling the arguments by real-valued factors. Fryer et al. (2019) modeled "double updating" when agents are faced with ambiguous information. The agent first interpret each signal based on their current belief and then form a posterior on the sequence of interpreted signals. In this chapter, however, belief polarization happens even with unambiguous information. This chapter also extends the idea of subjective perception of parameters/information by introducing a general perceptual set.

Three strands of the literature are particularly close to the mechanism developed here, and it is worth clarifying the differences. First, Ortoleva (2012) models an agent who updates by Bayes' rule until the received information is too surprising, and then switches to a different model of the world altogether, so that the posterior jumps discontinuously. There, the agent is aware that the model may be wrong and deliberately chooses a new one. In contrast, the perceptual set here is filtered through unconsciously: the agent never questions the quality of the sensory information, and revision is triggered endogenously by the accumulation of perceptual shocks through the entrenchment level, rather than by an active change of paradigm. Second, Rabin and Schrag (1999) show that confirmatory bias—misinterpreting ambiguous evidence as supporting the current hypothesis—induces persistent bias and overconfidence. My mechanism instead operates on continuous states through a perceived-signal transformation, and delivers persistent disagreement that can be narrowed or eliminated by set revision. Third, the family of diagnostic expectations (Bordalo et al., 2019, 2023) and motivated beliefs (Bénabou, 2015) derive over- and underreaction from representativeness and selective memory, respectively; here, the central result is a discontinuous revision of posterior beliefs when a perception map changes. Asset-price and volume implications require an additional equilibrium model.

2 Learning without perceptual bias

Although perceptual sets are generally shaped by personal traits, previous perceptual shocks and prior beliefs, they are very persistent. Once they are formed, the influence they receive from belief updating through combining new information is minimal. Revisions of perceptual sets occur mainly following huge perceptual shocks, which are unforeseeable rare events. In the absence of perceptual set revisions, the learning process can be modelled by optimal linear filtering. The agents are not aware of the perception process and bias, and the perceptual shocks are exogenous, unforeseeable and rare. Once perceptual shocks hit and the perceptual set is revised, the agent repeats the learning process and apply the optimal linear filtering under her new perceptual set. In this section, I introduce the general framework of optimal linear filtering,

which is joined piecewise once the perceptual set revision is induced exogenously.

Let (V_t, P_t), $t \in [0, T]$, be a two-dimensional process. The latent state V_t is not observed directly; P_t is a common raw signal:

$$\mathrm{d}P_t = A_t V_t \,\mathrm{d}t + B_t \,\mathrm{d}Z_t, \qquad \mathrm{d}V_t = a_t V_t \,\mathrm{d}t + b_t \,\mathrm{d}Z_t^f, \tag{1}$$

where A_t, $B_t > 0$, a_t, and b_t are deterministic and Z and Z^f are independent Brownian motions. The distinction between a raw signal and a perceived signal is essential. A perceptual set is a predictable process η_t^i, measured in units of signal drift, that maps common raw information into the signal processed by agent i:

$$\mathrm{d}\widetilde{P}_t^i = \mathrm{d}P_t + \eta_t^i \,\mathrm{d}t\,. \tag{2}$$

Thus agents share P, but generally do not share $\widetilde{P}^i$. Conditional on her perceived signal, each agent applies the same Bayesian filtering rule; the friction lies in the perception map, rather than in a departure from Bayes' rule. Let $\mathcal{F}_t^{\widetilde{P}^i}$ be the filtration generated by $\widetilde{P}^i$, and let $V_0 \sim N(m_0, \gamma_0)$. The conditional mean is summarized by the following proposition.

Proposition 1. *Let* $m_t^i \equiv \mathbb{E}\left[V_t \mid \mathcal{F}_t^{\widetilde{P}^i}\right]$ *and* $\gamma_t^i \equiv \mathbb{V}ar\left(V_t \mid \mathcal{F}_t^{\widetilde{P}^i}\right)$. *Then*

$$\mathrm{d}m_t^i = a_t m_t^i \,\mathrm{d}t + \frac{A_t \gamma_t^i}{B_t} \,\mathrm{d}I_t^i, \tag{3}$$

where the normalized perceived innovation is

$$\mathrm{d}I_t^i = \frac{1}{B_t}\left[\mathrm{d}\widetilde{P}_t^i - A_t m_t^i \,\mathrm{d}t\right] = \frac{1}{B_t}\left[\mathrm{d}P_t + \eta_t^i \,\mathrm{d}t - A_t m_t^i \,\mathrm{d}t\right]. \tag{4}$$

The conditional variance is deterministic and solves the Riccati equation

$$\mathrm{d}\gamma_t^i = \left(2a_t\gamma_t^i + b_t^2 - \frac{A_t^2(\gamma_t^i)^2}{B_t^2}\right)\mathrm{d}t, \qquad \gamma_0^i = \gamma_0. \tag{5}$$

In particular, $\gamma_t^i = \pi_t^i(V_t^2) - (m_t^i)^2$; it is never the first moment minus its square. A proof and the

constant-state solution appear in the appendix.

3 Learning under perceptual bias

3.1 Perceptual sets

There is a continuum of measure-one, ex-ante identical agents. Agent i has a perceptual set η^i, which determines the transformation in (2). This is not a second probability measure and does not alter the physical law of P. Instead, it captures the agent's unconscious conversion of a common sensory input into a private internal representation. A Bayesian benchmark has $\eta^i_t = 0$ at all dates. This formulation permits agents to be fully Bayesian after perception while locating the behavioral friction before belief updating.

3.2 Revision of perceptual sets

3.2 Revision events and set entrenchment

Let N^i_t be a Poisson counting process with intensity λ_i, independent of (Z, Z^f), and let $\{\beta^i_n\}_{n\geq 1}$ be independent marks in $[0,1]$. Its arrival times T^i_n are perceptual shocks. Between arrivals, entrenchment evolves as

$$\mathrm{d}E^i_t = \alpha_i \,\mathrm{d}Z_t, \qquad E^i_0 > 0. \tag{6}$$

An arrival lowers entrenchment by $\zeta_i \beta^i_n$, and the revision time is

$$\tau_i = \inf\{T^i_n : E^i_{T^i_n -} - \zeta_i \beta^i_n \leq 0\}. \tag{7}$$

At τ_i, the set is revised and entrenchment resets:

$$\eta^i_{\tau_i} = (1 - \beta^i_n)\eta^i_{\tau_i -}, \qquad E^i_{\tau_i} = E^i_0. \tag{8}$$

Full correction is the special case $\beta_n^i = 1$; otherwise revision attenuates rather than eliminates bias. At τ_i, the agent recomputes her posterior using the entire transformed history $\{\widetilde{P}_s^i : 0 \leq s \leq \tau_i\}$ under the revised set. Consequently, the posterior can jump even without a new raw signal. This reprocessing result is the mechanism studied below.

3.3 Results

In this section, I first present the special case where the state of the world does not change, and then a more general case with a stochastic world state.

3.3 Special case: time-invariant state of the world

Consider the special case where the unknown state of the world V is time-invariant, and the observable P_t is simplified to $\mathrm{d}P_t = V\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t$. Agent i processes $\mathrm{d}\widetilde{P}_t^i = \mathrm{d}P_t + \eta^i\,\mathrm{d}t$, where η^i is constant between revision events. The conditional moments follow from (3)–(5).

Proposition 2. *In the special case where $A_t = 1$, $B_t = \sigma$, and the state of the world V is time-invariant, consider the Bayesian agent and the behavioral agent in the absence of perceptual set revision.*

1) The time-t conditional mean of V from the Bayesian agent's view is

$$m_t = \frac{m_0\sigma^2 + P_0\gamma_0}{\gamma_0 t + \sigma^2} + V\frac{\gamma_0 t}{\gamma_0 t + \sigma^2} + \frac{\gamma_0\sigma}{\gamma_0 t + \sigma^2}\int_0^t \mathrm{d}Z_t\,. \tag{9}$$

2) The behavioral agent's view differs by

$$m_t^i = m_t + \eta^i\frac{\gamma_0 t}{\gamma_0 t + \sigma^2}. \tag{10}$$

3) All agents agree on the conditional variance of the underlying state:

$$\gamma_t^i = \gamma_t = \frac{\gamma_0\sigma^2}{\gamma_0 t + \sigma^2}. \tag{11}$$

4) *The Bayesian agent eventually learns about the true state of the world V, while behavioral agent i eventually thinks the state of the world to be $V + \eta^i$.*

In this special case, all agents, regardless of their perceptual sets, share the same conditional variance of the state variable, which is decreasing in t. The conditional expectations, however, vary with the perceptual bias. At a fixed time, perceptual shocks are imposed. The conditional moments can be calculated piecewise. After the perceptual shocks, agents still agree with the conditional variance of the state variable, the conditional expectation still varies by the remaining perceive bias.

If revisions completely correct perceptual bias, the conditional expectations of all agents coincide with the Bayesian posterior immediately after the revision. Partial revisions instead shrink, but do not generally eliminate, disagreement. As an illustration, Figure 2 plots the empirical distribution of the conditional expectation of the state variable and the empirical distribution of the entrenchment level at time T. In this special case, the true value of V is set to be 1, and the default entrenchment level is normalized to 1. Tests do not reject the normal distribution hypothesis for either panel.

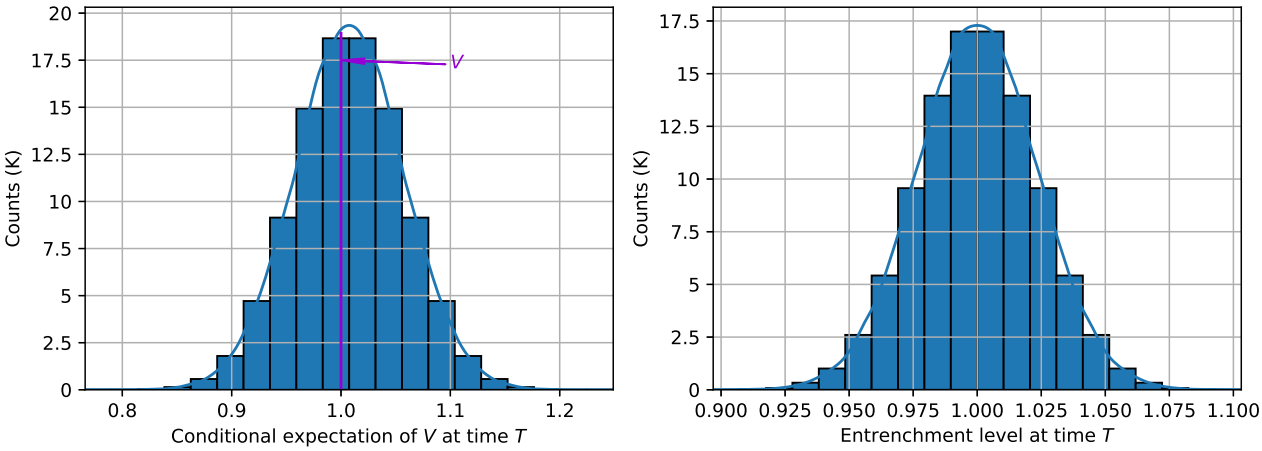


Figure 2. Distributions of conditional expectations of V and entrenchment levels. The left panel plots the empirical distribution of $m_T^i = \mathbb{E}^i[V \mid \mathcal{F}_T^\xi]$ with $V = 1$ and $\eta^i \sim N(0, 1)$. The true value $V = 1$ is also noted on the plot. The right panel plots the empirical distribution of E_T^i with the default entrenchment level, which is the level to which entrenchment adjusts after a perceptual set revision, normalized to 1. Tests do not reject the normal distribution hypothesis for either panel.

Figure 3 plots the random paths of the conditional expectation of the underlying state. Ab-

sent of perceptual shocks, the conditional expectations change smoothly from updating with new information. When exogenous perceptual shocks are introduced at a fixed time ($t = 10$), perceptual sets are revised and thus conditional expectations experience jumps. The magnitude of the jumps reflects the effects of perceptual set revision, isolated from any incoming new information. Assume the sensitivity to perceptual shocks and the intensity of perceptual shocks are large enough, then the perceptual bias is immediately eliminated. The conditional expectations of the underlying state by all agents converge to its true value.

3.3 General case

For a time-varying state, equations (3) and (5) characterize learning between revision times. At a revision time, the agent applies the same filter to the complete perceived history under the new perception map. This construction separates an ordinary innovation-driven update from a discontinuous posterior revision caused by a change in interpretation. The general case does not by itself determine an asset price, trading volume, or an overreaction statistic; those objects require an additional equilibrium model.

4 Connecting with the financial markets

The learning model is not, by itself, an equilibrium asset-pricing model. In particular, the common raw signal P_t in (1) is not identified with an equilibrium stock price. A market application requires a separate price process, investors' portfolio demands, a supply specification, and a market-clearing condition. We therefore treat the financial interpretation as a research extension rather than a result of the present model.

In such an extension, "overreaction" must be defined relative to a stated benchmark, for example the price response in the common-perception Bayesian economy. Likewise, a trading-volume implication requires equilibrium holdings and an aggregation rule. The current paper establishes the informational mechanism—a revision of perception can produce a discrete posterior change without new raw news—but does not claim an equilibrium price or volume result.

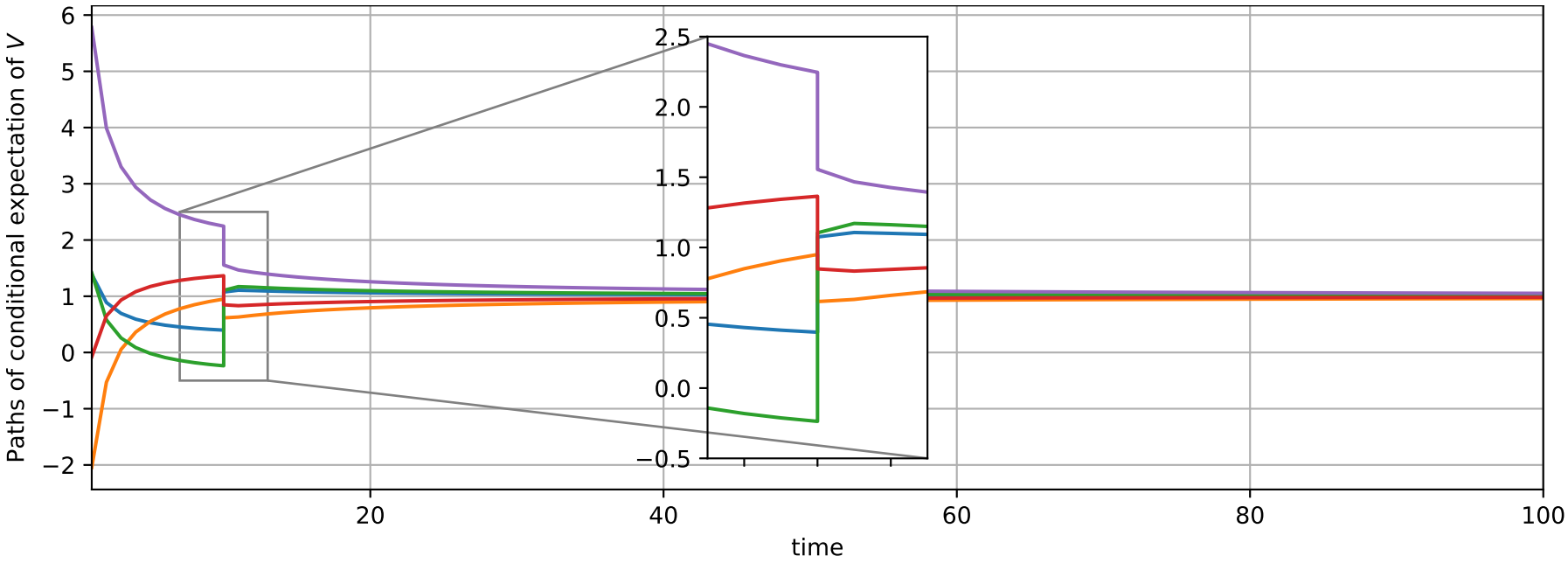


Figure 3. Paths of conditional expectations of θ with perceptual set revisions. Five random paths of m_t^i, the conditional expectation of the underlying state, are plotted. When exogenous perceptual shocks are introduced at time $t = 10$, perceptual sets are revised and thus conditional expectations experience jumps. The magnitude of the jumps reflects the effects of perceptual set revision, isolated from any incoming new information.

5 Conclusion

This chapter distinguishes raw sensory information from the perceived signals on which agents condition. Perceptual sets are maps from a common raw signal to agent-specific perceived signals; agents then use the same Bayesian filter on those perceived signals. Entrenchment makes these maps persistent, while marked Poisson shocks can trigger a revision and a reset of entrenchment.

The model generates persistent disagreement when perceptual sets differ. A revision changes the interpretation of the entire observed history and can therefore produce a discontinuous posterior change without new raw news. Full revisions eliminate the distortion; partial revisions reduce it. Asset-price overreaction and trading-volume implications require a separate equilibrium extension and are not claimed as results here.

The perceptual set mechanism can be applied to many other scenarios, for example, belief polarization in political economy. Traditionally, political polarization affects finance through two channels. Through the partisan utility channel, partisans obtain utility through from engaging with their fellow party members or from owning specific financial securities. My perceptual set mechanism is close to the partisan beliefs channel. Partisans may hold different beliefs regarding the state of the economy, expected risk-adjusted returns in the cross section of stocks, or the ability of in-group versus out-group partisans. The notion of a *partisan perceptual screen* was introduced in a seminal study by Campbell et al. (1960). Heterogeneity in beliefs could arise from differences in information environments (e.g., via partisan media), differences in attention, or differences in individuals' models of the world.

References

Emily Balcetis and David Dunning. See what you want to see: motivational influences on visual perception. Journal of Personality and Social Psychology, 91(4):612, 2006.

Roland Bénabou. The economics of motivated beliefs. Revue d'Économie Politique, 125(5):665–685, 2015.

Pedro Bordalo, Nicola Gennaioli, Rafael La Porta, and Andrei Shleifer. Diagnostic expectations and stock returns. Journal of Finance, 74(6):2839–2874, 2019.

Pedro Bordalo, John J Conlon, Nicola Gennaioli, Spencer Y Kwon, and Andrei Shleifer. Memory and probability. Quarterly Journal of Economics, 138(1):265–311, 2023.

Jerome S Bruner and A Leigh Minturn. Perceptual identification and perceptual organization. The Journal of General Psychology, 53(1):21–28, 1955.

Jerome S Bruner and Leo Postman. On the perception of incongruity: A paradigm. Journal of Personality, 18(2):206–223, 1949.

A Campbell, Converse PE, Miller WE, and Stokes DE. The American Voter. Wiley, New York, 1960.

Kyle Chauvin. Belief updating with dissonance reduction. 2023.

Hui Chen, Anton Petukhov, Jiang Wang, and Hao Xing. The dark side of circuit breakers. The Journal of Finance, 79(2):1405–1455, 2024.

Haw Cheng and Alice Hsiaw. Distrust in experts and the origins of disagreement. Journal of Economic Theory, 200:105401, 2022.

Marvin M Chun and Jeremy M Wolfe. Visual attention. Blackwell handbook of sensation and perception, page 272–310, 2005.

Pierre Collin-Dufresne, Michael Johannes, and Lars A Lochstoer. Parameter learning in general equilibrium: The asset pricing implications. American Economic Review, 106(3):664–98, 2016.

Jr Fryer, Roland G, Philipp Harms, and Matthew O Jackson. Updating beliefs when evidence is open to interpretation: Implications for bias and polarization. Journal of the European Economic Association, 17(5):1470–1501, 10 2019. ISSN 1542-4766. doi: 10.1093/jeea/jvy025. URL https://doi.org/10.1093/jeea/jvy025.

Sven Hansson. Stanford encyclopedia of philosophy: Logic of belief revision. https://plato.stanford.edu/entries/logic-belief-revision/, 2017. [Online; accessed 5-November-2019].

Scott Duke Kominers, Xiaosheng Mu, and Alexander Peysakhovich. Paying (for) attention: The impact of information processing costs on bayesian inference. Available at SSRN 2857978, 2018.

Korhan Koçak. Sequential updating: A behavioral model of belief change. Technical report, Working Paper, 2018.

Mark Krause, Daniel Corts, Stephen Smith, and Dan Dolderman. Psychological Science. Pearson Cananda, Don Mills, second edition, 2017.

Raymond S Nickerson. Confirmation bias: A ubiquitous phenomenon in many guises. Review of general psychology, 2(2):175–220, 1998.

Kristoffer Nimark and Savitar Sundaresan. Inattention and belief polarization. Journal of Economic Theory, 180:203–228, 2019.

Pietro Ortoleva. Modeling the change of paradigm: Non-Bayesian reactions to unexpected news. American Economic Review, 102(6):2410–2444, 2012.

Matthew Rabin and Joel L Schrag. First impressions matter: A model of confirmatory bias. Quarterly Journal of Economics, 114(1):37–82, 1999.

Leonard Savage. The Foundations of Statistics. Wiley, New York, 1954.

Daniel Schacter, Daniel Gilbert, Daniel Wegner, and Matthew Nock. Psychology, chapter 4. Worth, New York City, third edition, 2014.

Daniel J Simons and Christopher F Chabris. Gorillas in our midst: Sustained inattentional blindness for dynamic events. perception, 28(9):1059–1074, 1999.

Wayne Weiten. Psychology: Themes and variations. Wadsworth/Thomson Learning, 2001.

Appendix

A General framework of optimal linear filtering

For completeness, this appendix records the filter used in the main text. Under (1) and (2), the perceived observation has the form

$$\mathrm{d}\widetilde{P}_t^i = (A_t V_t + \eta_t^i)\,\mathrm{d}t + B_t\,\mathrm{d}Z_t\,.$$

Agent i treats $\widetilde{P}^i$ as her observed signal and uses the linear-Gaussian Kalman–Bucy filter. Writing

$$\mathrm{d}I_t^i = B_t^{-1}\left(\mathrm{d}\widetilde{P}_t^i - A_t m_t^i\,\mathrm{d}t\right),$$

the posterior mean and variance satisfy

$$\mathrm{d}m_t^i = a_t m_t^i\,\mathrm{d}t + \frac{A_t\gamma_t^i}{B_t}\,\mathrm{d}I_t^i, \qquad \dot{\gamma}_t^i = 2a_t\gamma_t^i + b_t^2 - \frac{A_t^2(\gamma_t^i)^2}{B_t^2}.$$

The innovation is normalized by B_t, which is why the gain includes the factor $1/B_t$. These equations also imply $gamma_t^i = \pi_t^i(V_t^2) - (m_t^i)^2$.

For the constant-state case, $A_t = 1$, $B_t = \sigma$, and constant η^i,

$$\gamma_t = \frac{\gamma_0\sigma^2}{\gamma_0 t + \sigma^2}, \qquad m_t^i = \frac{m_0\sigma^2 + \gamma_0(P_t - P_0 + \eta^i t)}{\gamma_0 t + \sigma^2}.$$

Since $P_t - P_0 = Vt + \sigma Z_t$, this gives

$$m_t^i = \frac{m_0\sigma^2}{\gamma_0 t + \sigma^2} + (V + \eta^i)\frac{\gamma_0 t}{\gamma_0 t + \sigma^2} + \frac{\gamma_0\sigma}{\gamma_0 t + \sigma^2}Z_t.$$

Thus, between revisions, agents with different perception maps can hold different posterior means despite sharing the same raw signal. A full revision sets $\eta^i = 0$ and aligns the posterior with the Bayesian benchmark; a partial revision narrows disagreement without necessarily eliminating it.